Case H — Mr. David Thompson | 55M | Chronic Bronchitis (Type B COPD — "Blue Bloater")

Bus driver. Productive cough ≥3 months/year for | 2+ consecutive years. Bluish lips, bilateral peripheral oedema lower

Target: 70%+ | Cardiorespiratory Physiotherapy

■ PROBLEM LIST

- Excess mucus production — impaired mucociliary clearance
- Airflow obstruction (reduced FEV1/FVC on spirometry)
- Chronic hypoxaemia and possible CO2 retention
- Cor pulmonale / right heart strain (oedema, cyanosis)
- High BMI increasing work of breathing
- Active smoking perpetuating airway inflammation
- Recurrent sick leave — primary presenting concern
- Reduced exercise tolerance and exertional dyspnoea
- Prolonged sitting at work worsening dependent oedema
- Possible DVLA fitness-to-drive consideration
- Nicotine dependence as a behavioural barrier
- Sleep disruption from nocturnal cough / possible OSA (high BMI)

■ RED FLAGS

None identified

■ PATHOPHYSIOLOGY

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■ DIFFERENTIAL DIAGNOSES

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■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

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■ TREATMENT PLAN

■ Airway Clearance: Active Cycle of Breathing Technique (ACBT)

- Full technique:
- BC (4–6 relaxed breaths) →
- TEE (3–5 deep breaths with 3-second inspiratory hold) → FET/Huff (1–2
- huffs at mid-volume, 1 at high volume) → BC. Repeat 3–4 cycles. Perform
- upright seated or in high side-lying. Aim for twice daily — morning on

■ Breathing Exercise: Pursed-Lip Breathing (PLB)

- Breathe in through the nose for
- 2 counts. Breathe out slowly and gently through pursed lips for 4
- counts. PLB maintains positive pressure in the small airways on
- exhalation, preventing dynamic collapse — particularly important in
- obstructive disease where airways are prone to collapse during forced

■ Exercise & Mobility: Graded Walking Programme

- Starting level:
- Begin at a pace
- maintaining Borg ≤4/10 and SpO2 ≥88%. For a high-BMI patient with likely
- moderate COPD, target 10–15 minutes of continuous walking at a
- comfortable pace. Time the session to follow morning ACBT and inhaler

■ Smoking Cessation — Motivational Interviewing Approach

- Assess readiness:
- "On a scale of 0 to
- 10, how important is it to you to quit smoking, and how confident are
- you that you could if you decided to?" Explore what would shift the
- number upward — identify ambivalence, not resistance.